Resampling Structured Data: Recap

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Models say what to resample

The day 1 bootstrap assumes the observations are iid. When they are not, use a model to say which parts of the data are iid, and resample those.

Model	Resample
X_t iid	observations
random walk, $X_t = X_{t-1} + \epsilon_t$	differences $X_t - X_{t-1}$
trend plus stationary noise, $X_t = m(t) + Z_t$	blocks of residuals
$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, ϵ_i iid	residuals
the same, with $\mathrm{Var}(\epsilon_i)$ depending on x_i	residuals with random signs
(X_i, Y_i) iid	rows
groups with their own distributions and fixed sizes	rows within each group

Dependent data: Lake Huron

The slope of the lake's level on year, 1875–1972, is -0.0242 feet per year. The SE from `summary` is 0.0040 and assumes independent residuals.

- **Random walk.** Resample the differences and add them up from x_1. The bootstrap series wander off and stay there, while Lake Huron keeps returning to its level. The model is wrong, and its SE is 20 times the one from `summary`.
- **Trend plus stationary noise.** $X_t = m(t) + Z_t$, where Z_t is **stationary**: the same distribution in every year, with correlations that depend only on the lag.
- The residuals have lag-1 autocorrelation 0.76, and the autocorrelations are near 0 by lag 5. Resampling single residuals breaks that dependence.

The moving block bootstrap

1. Fit the trend and keep the residuals $z_1, \ldots, z_n$.
2. Draw blocks $(z_j, \ldots, z_{j+\ell-1})$ with replacement from the $n - \ell + 1$ starting points, put them end to end, and keep the first n values.
3. Add them to the fitted trend, $x_t^* = \hat{m}(t) + z_t^*$, and refit the slope.

block length ℓ	1	2	5	10	20
SE of the slope	0.0040	0.0053	0.0070	0.0077	0.0077

- **Failure case:** $\ell = 1$ resamples single residuals. It agrees with `summary`, and both understate the SE.
- Increase ℓ until the SE levels off. The best ℓ for an SE grows like $n^{1/3}$, but the constant depends on the dependence.
- With $\ell = 10$ the normal interval is [-0.0392, -0.0092].

Regression: three schemes

With $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, the fit gives fitted values $\hat{y}_i$ and residuals e_i. Each e_i^* is drawn from the residuals, and each V_i is ± 1 with probability $1/2$.

scheme	bootstrap sample	assumes	`cars` SE
cases	rows (x_i, y_i)	the pairs are iid	0.4122
errors	$y_i^* = \hat{y}_i + e_i^*$	the errors are iid	0.4077
wild	$y_i^* = \hat{y}_i + e_i V_i$	the fitted line is right	0.3995
`summary`		iid normal errors	0.4155

- Errors and wild keep the original x values. Cases refit with random x.
- Plot the residuals against x before resampling errors.

Why the wild bootstrap

When $\text{Var}(\epsilon_i)$ depends on x_i, resampling errors moves each residual away from the x it came from.

- The extreme x values have the most influence on the slope. If their errors are the largest, resampling errors understates the SE; if they are the smallest, it overstates it.
- In the wild bootstrap y_i^* has mean $\hat{y}_i$ and variance e_i^2, so the spread at each x_i comes from the residual at x_i.
- One e_i^2 is a poor estimate of $\text{Var}(\epsilon_i)$, but the slope combines all n of them.
- For `cars` the spread grows with speed, but not enough to change the SE.

Strata

`state.x77`: the percent of adults with a high school diploma in each of the 50 states, which fall into four regions of fixed sizes.

interquartile range, 11.1	SE
resample within each region	2.34
resample all 50 states	2.99

- Resampling all 50 lets the number of states from each region vary. The South is low and the West high, so the mix adds variation that is not in the data.
- Resample within each group when the group sizes are fixed, or when the statistic compares groups. The West's median minus the South's is 20.9 points, with percentile interval [12.6, 22.9].